cw 2 very good

Machine Learning for Computer Vision
Report for Coursework on image matching

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Abstract

The report is the basic concept of machine learning for computer vision, and implementation of some methods and approaches such as Homography and Fundamental Matrix as well as stereo vision, and other interesting findings.

I. Matching

a) Automatic

1) Interest Point Detector

In this part, we introduce Harris Interest Point Detector. According to C. Harris et al [1], Harris algorithm finds interest points and corners via detecting local maximum depending on a certain threshold. Parameters such as α is the corner response function, matrix for derivatives and convolution, matrix for blurring, and threshold. For every single experiment, proper parameters should be chosen for a good performance.

Table1 Number of Interest Points in Tsukuba Image

Interest Pts Number	Threshold =3e7	Threshold=3e8
α=0.01	1316	553
α=0.1	895	344

As we can see, as the value of threshold and α increases, the number of interest points decreases, which can also be verified by Figure1 below.



Fig.1 interest points detection (a) threshold=3e7 (b) threshold=3e8

2) Descriptor Generator

In this part, the descriptors of an interest point are generated simply by computing the color histogram of the 32x32 patch with the dimensionality of M=128. An interest point is the center of a patch and the value of each element in the histogram reflects different the color depth. Thus, a descriptor obtained can be characterized by a 128 by 1 matrix.

Some other useful and effective method like SIFT and SURF etc offers good detection and matching result [2]. It can be implemented using a mature toolbox and is fast and compact as well as good generalization ability. In the following experiment, the two methods are both implemented while only the results from color histogram are shown (result from SIFT in the appendix).

3) Neighbour Matching

The neighbor matching method that implemented is to find the pair of descriptors in two images with the minimum Euclidean distance between the descriptor and define them as the corresponding point pairs. The whole process is considered a matching. Some other distance can also be used, such as Mahalanobis distance etc. Figure2 below shows some pairs of one matching result that between two similar images using Euclidean distance, each line links two matching points. What can be found in the Figure2 is that the matching has quite satisfying performance and these pairs shown below are of high correspondence.



Fig.2 Matching result of some points

b) Transformation estimation

1) Homography Matrix (H)

It is credible to say that any two 2-D images of some correspondence are related by a homography matrix. The homography matrix is estimated by solving $X_2 = H \cdot X_1$, where X_1, X_2 are coordinates of two set of point pairs. In specific, the smallest column of V is reshaped to get H. V is obtained through the singular value decomposition of the matrix A, where

$$A = [a_{x1} \quad a_{y2} \quad \cdots \quad a_{x_N} \quad a_{y_N}]^T \tag{1}$$

where a_{xl}, a_{yl} denote certain combination of coordinates (refer to Appendix I). The ideal and practical homography matrix H corresponding to the two images in Figure2 are

$$H_{ideal} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad H_{practical} = \begin{bmatrix} 1 & 0 & -4.2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

The practical homography matrix is quite close to the ideal one, with only a bias of 0.2.

2) Fundamental Matrix (F)

The fundamental matrix describes a relation between two images of the same scene that some interest points appear in both images. The approaching of the fundamental matrix involves decomposition of matrix $A = USV^T$, where

$$A = [x_1x_2 \quad y_1x_2 \quad x_2 \quad x_1y_2 \quad y_1y_2 \quad y_2 \quad x_1 \quad y_1 \quad 1] \tag{3}$$

where x_l, y_l denotes the vectors of x and y coordinates. And then we use the smallest column of V to get F and update F via some tricks (refer to Appendix I). Sometimes we need to introduce a normalized algorithm to handle the issue of ill-condition for F. The steps of acquiring F is also shown in Appendix I. For example, the following matrix F is the fundamental matrix we get in the experiment,

$$\mathbf{F} = \begin{bmatrix} -6.72 \times 10^{-7} & 3.02 \times 10^{9} & -0.18 \\ -3.02 \times 10^{9} & -5.87 & 6.62 \times 10^{14} \\ 0.04 & 6.61 \times 10^{14} & 1 \end{bmatrix}$$

Fundamental matrix can be utilized for visualizing epipolar line and epipoles, which reflects the parameter of the environment such as baseline and camera center etc.

3) Projection and Homography Accuracy (HA)

By relocating every pixel from the original image, a projection process is achieved and a projected image can be

Summary of Comments on \\icnas2.cc.ic.ac.uk\ys316 \Desktop\MachineLCW\CourseWorkII\Reports\Machine Learning for Computer Vision

Page: 1

Number: 1 Author: . Subject: Sticky Note Date: 18/05/2017 17:29:46
General: good formatting. Extended results in the appendix, good. Some citations. Conclusion included.
1) Shows implementation, with some parameters and interest points values. Uses color histogram with SIFT results on appendix. Shows NN results. An little bit of discussion of the drawbacks of color histogram/SIFT could have been a nice extra.
2) Implements homography matrix estimation and shows results compared to a known matrix, good. Tests HA using a separate test set, which is a nice touch (standard deviation added). Shows epipolar lines in translation, makes sense.
3) Good indepth analysis, visually and numerical. HA compared, with STD also. Homography parameters found and well explained. Good plot of HA depending on number of correspondences and use of RANSAC and good analysis.
4) Well done. Disparity map explained, depth map consistent, focal distance change explained with formula, noise results good, rectification done.

Number: 2 Author: . Subject: Sticky Note Date: 18/05/2017 17:17:49
Shows matching example. Good.

Number: 3 Author: . Subject: Sticky Note Date: 18/05/2017 17:18:17
Good, own implementation of Harris. It shows example of the algorithm and performance depending on the parameters.

Number: 4 Author: . Subject: Sticky Note Date: 18/05/2017 17:22:40
Confusing sentence, not clear the term some correspondence.

Number: 5 Author: . Subject: Sticky Note Date: 18/05/2017 17:18:53
Table could have been formatted better. It seems that \alpha is the interest points number.

Number: 6 Author: . Subject: Sticky Note Date: 18/05/2017 17:22:51
Well done showing a quick example on a known pair of images.

Number: 7 Author: . Subject: Sticky Note Date: 18/05/2017 17:24:35
Quick analysis of color histogram drawbacks against SIFT, etc could have been good. Also maybe showing patches and its descriptors.

Number: 8 Author: . Subject: Sticky Note Date: 18/05/2017 17:24:46
Experiment done with set of images FD?

visualized. Some blemishes can be removed by averaging surrounding pixels to get a smooth figure, especially for a case of zooming in, which is considered to be spatial masks.

As the original interest points in a certain image are known, the Homography Accuracy is reasonable to be calculated by these interest points that known and those projected from another image. Obviously, point pairs that participate in obtaining H should not be used for measuring HA. So during the process, up to 80% corresponding interest point pairs are utilized for acquiring H, and the at least 20% remained to take part in testing HA.

Table2 HA between Tsukuba1 and other Tsukuba scenes

Images	Tsukuba2	Tsukuba3	Tsukuba4	Tsukuba5
HA	2.12	2.34	2.40	2.62
STD	0.55	0.54	0.53	0.51

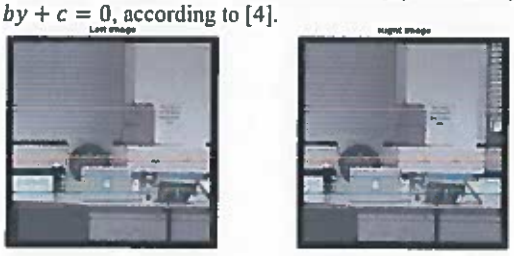
For those cases that the projection from Tsukuba1 to another is satisfying, Table2 shows HA between them. Those projection failure cases are not suitable for HA calculation. The interest points used are chosen automatically and the HA is within 3, which means a fair good projection with a limited error.

4) Epipolar Line

Epipolar plays an important role in image processing such as matching and reconstruction. The two points of matching pairs must lie on the same epipolar line in each image. And every epipolar line comes from each image must go through the same point which is called epipole [3]. Thus a matching or a stereo rectification of two similar images of the same scene may be done using epipolar lines and poles, which is 1-D matching instead of 2-D for the original image and thus makes things simpler. The epipolar line is introduced based on the fundamental matrix obtain in part I.b.2. The main idea is to solve

$$[a\ b\ c]^T = F \cdot [x\ y\ 1]^T \qquad (4)$$

where x, y denote the vectors of x and y for all interest point pairs and parameter a, b, c determine a epipolar line by $ax + by + c = 0$, according to [4].



(a) (b)

Fig.3 Epipoles and Epiplar lines (a) left image (b) right image

Figure3 shows some of the epipolar lines going through corresponding interest points of the two images. Consider that the movement of the camera is quite small, the translation can be regard as being parallel to the image plane and without a difference in look angle. As the epipolar lines are parallel as shown above and epipoles are at infinity.

II. Image Geometry

a) Homography

1) HA Comparison

Table3 Averaged Homography Accuracy between Original and resized one by factor 2, using manually chosen interest points pairs, harris thr=3e8

Images	Lab	Boat	Tsukuba	NotreDamedeParis
HA	4.82	5.67	5.21	5.13
STD	1.18	1.54	1.53	1.29

Table4 Averaged Homography Accuracy between Original and resized one by factor 2, using automatically chosen interest points pairs, harris thr=3e8

Images	Lab	Boat	Tsukuba	NotreDamedeParis
HA	2.11	2.64	2.30	1.90
STD	0.65	0.54	0.53	0.51

In this part, we assume to have an ideal homography matrix that represents the transformation of reducing the size. So s would be 0.5 and the α would be 0. Then the interest points in original image are projected using H and distance between these projected points and those detected in small size image. As we can see in the Table4, HAs are about 2.5 when using automatically chosen points pairs and about 5 when using manually chosen pairs. Standard deviation shows limited data fluctuation and using automatic correspondences are more reliable. All the cases have acceptable low HA values.

The number of points may change because it is determined by threshold and other parameters of the detector and the image as well. If we properly control the number of interest points for both two images, it can be found that the Homography approaches still apply after reducing size according to the low level of HA obtained above. This is because that the interest points of the original image, such as a corner, also appear after reducing the size of the image. It means reducing size do not effect on the detection of certain interest point.

2) Visualization and Validation of Homography and Parameters



(a)Original Image (b) Rotated and zoomed out Image

Fig.4 Original Image and Parallel Motion Image

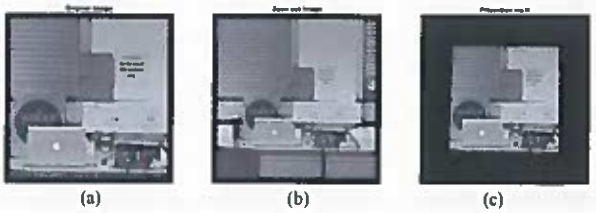
Figure 4 shows the original image and the image rotated and zoomed out, we use in this part. Different correspondence is chosen to implement the projection.



(a) (b) (c)

Fig.5 Projected Image using (a)automatic correspondences(b)carefully chosen correspondeces (c)less accuracy manually chosen correspondeces

What can be found in the Figure5 is that manually chosen points pairs can not guarantee a good performance due to some blemishes and deformation are shown in Figure5(c). However, automatic approach circumvents the problem due to its reliability in detecting interest point. To get more suitable pairs without incorrect match points, an algorithm called 'ransac' is introduced and will be discussed later.



(a) (b) (c)

Fig.6 Zoomed out Image Projection (a)Original Image (b)Zoomed out Image (c)Projection Image

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Number: 1 Author: . Subject: Sticky Note Date: 18/05/2017 17:27:07
Ok, good.

Number: 2 Author: . Subject: Sticky Note Date: 18/05/2017 17:24:58
This is right.



(a) (b) (c)

Fig.7 Rotated Image Projection (a)Original Image (b) Rotated in Image (c)Projection Image

Consider a general condition that the geometric transformation may contain rotation, similarity as well as translation. All the transformations can be characterized by the value of H. In specific, the H obtained in Figure7 is,

$$Homography\ Matrix = \begin{pmatrix} 0.986 & 0.217 & -60.1 \\ -0.217 & 0.986 & 63.4 \\ 0 & 0 & 1 \end{pmatrix} \quad (5)$$

And a general homography matrix is

$$General\ HM = \begin{pmatrix} s\cdot\cos(\alpha) & s\cdot\sin(\alpha) & t_x \\ -s\cdot\sin(\alpha) & s\cdot\cos(\alpha) & t_y \\ 0 & 0 & 1 \end{pmatrix} \quad (6)$$

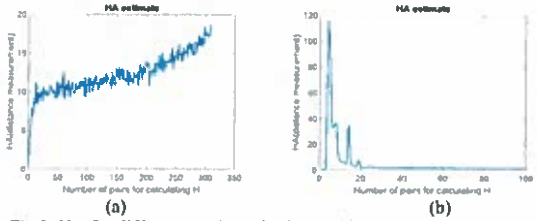
The length ratio s and the angle α are derived from equation (5) and (6) that $(s\cdot\sin(\alpha))^2 + (s\cdot\cos(\alpha))^2 = 0.986^2 + 0.217^2 = 1$, so $s=1$ and $\alpha \approx 12.5°$. As a result, the transformation is to rotate the image anti-clockwise for 12.5 and without zooming.

Table5 parameters of some transformation, harris detector with thr=3e8

	α	s	t_x	t_y
Translation	0.7	1.00	50.5	1.7
Zoom out	-0.8	0.66	108.2	109.8
Rotation	12.5	1.00	-60.1	63.4

Compared with rotation case discussed before, translation case has nearly no rotation and zoom because the α is only 0.7 and $s = 1$. And the X-coordinate has increased by 50.5 but Y-coordinate slightly changed, which means the image has mainly a parallel motion by about 50 in X-coordinate. Likewise, the zoom out case does not involve translation or rotation, but only zooms out by a scale of 0.66. Some projection results using other interest point detectors are shown in the Appendix III.

3) HA and Outliers



(a) (b)

Fig.8 HA for different number of pairs (a) without ransac (b) ransac

The parallel motion image and its original one are used in this part. According to Figure8 (a), HA growths when the number of correspondences increases. This is because some detected interest points are matched by mistake and when the number of chosen pairs are over a certain level, it surely contains some bad pairs, which lead to the poor estimation of H.

To solve the problem, RANSAC [4] is introduced to discard unsuitable pairs. Random sample consensus is an iterative method that detects outliers with no effect on the value of estimation homography matrix, and the main algorithm is,

1. Find interest points and pairs with high correlation
2. Four points are selected from the set to get a homography.
3. Pairs agreeing with $dist(Hx, x') < threshold$ are selected.
4. Repeat steps 3 and 4 until a sufficient number of pairs meet the requirement in 3.

After applied RANSAC, it can be found in Figure8 (b) that the HA obtained declined as the number of pairs used increases, which corresponds to general cognition that more pairs provide more information thus leading to satisfying accuracy.

b) Stereo Vision

1) Fundamental Matrix Estimate

In this part, we estimate fundamental matrix of other images, which is shown in the next part. The F obtained using manually chosen(F_m) points pairs and automatic ones (F_a) are below,

$$F_m = \begin{pmatrix} 0 & 0 & 0.07 \\ 0 & 0 & -0.20 \\ -0.07 & 0.18 & 1 \end{pmatrix} \quad F_a = \begin{pmatrix} 0 & 0 & 0.08 \\ 0 & 0 & -0.17 \\ -0.07 & 0.16 & 1 \end{pmatrix} \quad (7)$$

These two matrices are quite similar with each other, so F_a is chosen for visualise epipoles and epipolar lines next.

2) Epipoles and Epipolar Lines/Epipoles



(a) (b)

Fig.9 Epipoles and Epiplar lines (a) left image (b) right image

Before in partI.a.4, the epipoles are at infinity, and epipolar lines are parallel. In Figure9, the epipoles are still invisible in the image. But the epipolar lines are not parallel anymore. They would go through the same point somewhere outside the image. As is mentioned before that the two points of matching pairs must lie on the same epipolar line in each image, so epipolar lines and epipoles can be used for matching images by making epipolar lines in alignment. The method applies even for the parallel case since the position of lines is the same because of their correspondences.

3) Disparity Map

Disparity map determined by the pixel difference, which means the motion between a pair of the stereo image [5]. It is clear that if the camera moves a little, the objects that far away can be seen as fixed, or moving very little. However, those that are close to the lens will have an apparent change in position. This change can be captured by disparity map and it offers a figure of difference shades



(a) (b)

Fig.10 (a) original image (b) disparity map

In Figure10 (a), a stereo image is shown by overlaying one to the other directly. The red image represents left while blue comes from the right image. Figure10 (b) is the disparity map. It can be found in the disparity map that the objects that are close to the camera will be brighter while those further seems to be darker. The interesting thing is that the surface of the desk changes its shades smoothly from bri[illegible]ark, which verifies the reliability of disparity map.

To reduce the influence of noise, block size should not be too small. However, over large block size also decays the result because it discards considerable information. Disparity map range determines the quantitative classification of disparity and the moderate range is suitable since small range leads to insufficient in the result and the large range is not necessary. Changing in focal length would be unde[illegible]ssion later.

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Number: 1 Author: . Subject: Sticky Note Date: 18/05/2017 17:27:14
Really nice analysis, visually and numerically.

Number: 2 Author: . Subject: Sticky Note Date: 18/05/2017 17:27:42
Good, results of the disparity map explained.

Number: 3 Author: . Subject: Sticky Note Date: 18/05/2017 17:27:42
Good analysis of the block size.

4) *Depth Map*

A depth map is a 3D image that contains information of distance, considering the position of an object especially its surface. The depth map derives from a disparity map [6]. In the process, quantification leads to surface in short distance range showing a generally same color and depth. Thus depth map offers an embossment with one object appears at the same depth, being insensitive to the slight change in distance.

However, the limitation is inevitable. It can be less reliable when the image is more complex. Besides, noise effects on the performance of a depth map as well.



(a) (b)

Fig.11 Depth map of two different image(a) tsukuba (b) desk and cups

In Figure11, (a) shows the depth map of Tsukuba. In a depth map, the closer to lens an object is, the more bulged the part is, which can be proved compared with the original image in part I, a, 1. Figure11 (b) corresponds to the disparity map in part II, b, 3. It can be found that the mug is close to the camera and paper cup is further and the large box is behind the paper cup while the background plate in the end. The distance and position relation is correctly reflected by depth map. However, the small box behind the paper cup cannot be observed because its location in pixels is considerably near to paper cup, which makes it hard to detect.

5) *Change Focal Distance*



(a) (b)

Fig.12 Change focal length(a) original image (b) disparity map

Figure12 (a) (b) are original image and disparity map after increasing the focal length for the same scene in the previous part. The difference in shade between the mug and the box becomes significantly more obvious because the objects close to the camera move more obviously. So increasing the focal length makes the system even more sensitive to things in the front. As a result, the chromatism seems to be enlarged.



(a) (b)

Fig.13 Increase focal length (a) the clear image (b) image with maximum 2 pixel corrupted by random noise of 0.1 σ^2 (c) noise of 0.2 σ^2

In Figure13 (a), the change of focal length does not lead to an obvious difference in depth map but only a little increase in the difference between background plate and objects. This is because the depth raises up when focal length increase, which is determined by the following equation according to [6],

$$depth\ z = \frac{focal\ length * base\ line}{disparity} \quad (7)$$

The position relation between objects remains correct it is in Figure11 (b). The depth map offers information of distance as expected.

However, compared with the depth map without noise, the one with some small noise provides a not satisfying result because the depth is less clear than before depending on noise power. The total depth seems to be suppressed and the difference in depth and color become smaller, which makes it hard to observe. In Figure13 (b), when noise if of 0.1 power, the depth map shows narrower in total depth according to the change of color, however when noise power increase to 0.2 in Figure13 (c), the distance of objects become difficult to distinguishing. If the noise becomes even larger, the depth map fails to offer any distance information. This is because tradition depth map owns limited robustness and can be affected by noise easily.

6) *Rectification*

Rectification means to combine images onto the same plane, making the corresponding y-coordinates in alignment [7]. As a result, the process provides an image contains the left and right image, which is useful for 3D image reconstruction etc. Steps of Rectification is:

1. Find interest points and match interest points pairs.
2. Find inliers of correspondences using such as 'ransac'.
3. Calculate rectification transformations.
4. Construct rectification image.



(a) (b)

Fig.14 Rectification(a) Inliers of two images (b) Rectification result

Figure14 shows the inlier points of two images and the rectification result, which represents a satisfying of image matching.

Conclusion

In the experiment, Harris detector and color histogram as well as SIFT are used for finding and matching interesting points. Correctly implemented projection using Homography leads to satisfying HA and parameters of H represents transformation relation. Fundamental Matrix characterizes the relation between two images of the same scene and helps to get epipoles and epipolar lines. Disparity map and depth map offers information of distance but is easily affected by noise. Rectification provides a stereo view using interesting points and Fundamental Matrix.

Reference

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[2]D. Lowe, "Distinctive Image Features from Scale-Invariant Keypoints", International Journal of Computer Vision, vol. 60, no. 2, pp. 91-110, 2004.

[3]R. Bolles, H. Baker and D. Marimont, "Epipolar-plane image analysis: An approach to determining structure from motion", International Journal of Computer Vision, vol. 1, no. 1, pp. 7-55, 1987.

[4] Vincent, Etienne, and Robert Laganiére. "Detecting planar homographies in an image pair." Image and Signal Processing and Analysis, 2001. ISPA 2001. Proceedings of the 2nd International Symposium on. IEEE, 2001.

[5]N. Qian, "Binocular Disparity and the Perception of Depth", *Neuron*, vol. 18, no. 3, pp. 359-368, 1997.

[6]K. Mikolajczyk, "Machine Learning for Computer Vision - Geometry of Vision", Imperial College, London, 2017.

[7]R. Hartley and A. Zisserman, Multiple view geometry in computer vision, 1st ed. Cambridge, UK: Cambridge University Press, 2003.

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Number: 1 Author: . Subject: Sticky Note Date: 18/05/2017 17:28:58
Good analysis of the noise effect.

Number: 2 Author: . Subject: Sticky Note Date: 18/05/2017 17:29:11
The explanation is correct.

Number: 3 Author: . Subject: Sticky Note Date: 18/05/2017 17:28:43
Focal length

Number: 4 Author: . Subject: Sticky Note Date: 18/05/2017 17:27:53
Results are consistent.

Number: 5 Author: . Subject: Sticky Note Date: 18/05/2017 17:28:49
Well done using the formula to argue the results.

Appendix I

Implement of Homography Matrix (H)

According [1], to solve $X_2 = H \cdot X_1$,

we mean to solve

$$a_x^T h = 0$$

$$a_y^T h = 0$$

where $h = [H_{11}, H_{12}, H_{13}, H_{21}, H_{22}, H_{23}, H_{31}, H_{32}, H_{33}]^T$

$$a_x^T = [-x_1, -y_1, -1, 0, 0, 0, x_2'x_1, x_2'y_1, x_2']^T$$

$$a_y^T = [0, 0, 0, -x_1, -y_1, -1, y_2'x_1, y_2'y_1, y_2']^T$$

As a result, we form a matrix A,

$$Ah = 0$$

where

$$A = \begin{bmatrix} a_{x1}^T \\ a_{y1}^T \\ \cdots \\ \cdots \\ a_{xN}^T \\ a_{yN}^T \end{bmatrix}$$

Then the decomposition is applied to A,

$$A = USV^T$$

The smallest column of V is reshaped to get H.

Implement of Fundamental Matrix (F)

According to [2], the process of obtaining Fundamental Matrix is similar to Homography. The first step is to form matrix A,

$$A = [x_1x_2 \quad y_1x_2 \quad x_2 \quad x_1y_2 \quad y_1y_2 \quad y_2 \quad x_1 \quad y_1 \quad 1]$$

where x_i, y_i denotes the vectors of x and y coordinates. Then we do the decomposition and get

$$A = USV^T$$

The smallest column of V is reshaped to get F. And the decomposition is applied again to F,

$$F = U_1 D {V_1}^T$$

We update D by set the smallest singular value to zero and get a new D, then

$$F_{updated} = U_1 D_{updated} {V_1}^T$$

Reference

[1]"8 Point Algorithm Essential and Fundamental Matrices", Brigham Young University, Utah, United States, 2011.

[2]Agarwal, Anubhav, C. V. Jawahar, and P. J. Narayanan. "A survey of planar homography estimation techniques." Centre for Visual Information Technology, Tech. Rep. IIIT/TR/2005/12 (2005).

Appendix II

Some important function:

Initialize/ add path and read images in advance.

```
addpath('./src');
addpath('./src/ransac_homography');
addpath('./src/modifit');
addpath('./figures/tsukuba');
tsukuba=imread('truedisp.row3.col3.pgm
');
...
```

Harris detector:

```
function y=fHarrisDetector(x,varargin)
% usage ---------------------------
y=fHarrisDetector(x,threshold,filter)
% g default=fspecial('gaussian',[7
7],1);
% x is input image matrix
% y is out put coordinates
% threshold default=1000
% tsukuba
%--------------------------------
%   Yining Shen
%   Imperial College London
%   Electrical&Electron2c Eng2neering
%   1.April.2017
%--------------------------------
if length(varargin)==1
   g = fspecial('gaussian',[7 7],1);
   threshold=varargin{1};
end
if length(varargin)==2
    threshold=varargin{1};
    g=varargin{2};
end
size =3;
dx = ones(3,1)*[-1 0 1];
dy = dx.';
% g=ones(7,1)*[0.03 0.105 0.222 0.286
0.222 0.105 0.03];
% g = fspecial('gaussian',[7 7],1);
% tsukuba=double(tsukuba);
Ix = conv2(double(x),dx,'same');
Iy = conv2(double(x),dy,'same');
Ix2 = conv2(Ix.^2, g,'same');
Iy2 = conv2(Iy.^2, g,'same');
Ixy = conv2(Ix.*Iy,g,'same');
harris = (Ix2.*Iy2-Ixy.^2)-
0.1*(Ix2+Iy2+eps).^2;
mx = ordfilt2(harris , size.^2 ,
ones(size,size) );
harris = (harris==mx) & (harris >
threshold );
[row,col]=find(harris);
figure
imshow(x)
hold on
plot(col,row,'r.','linewidth',2);
y(1,:)=col;
y(2,:)=row;
end
```

Generate Descriptor using colour histogram

```
function
color_hist=fGenerateDescriptor(y,image
)
col1=y(1,:);
row1=y(2,:);
count=1;
for i=1:size(y,2)
    sel=floor(image(row1(i)-
15:row1(i)+16,col1(i)-
15:col1(i)+16)/2)+1;

color_hist(count,:)=histcounts(sel,[1:
129]);
    count=count+1;
end
end
```

Homography estimate

```
function
H=fHomographyMatrix(coor_in,coor_out)
%   ----------------------2nput-------
%   coor1 & coor1 are correspond2ng
coordinates and should be 2*N
%   coor1=H*coor2
%   from coor2 to coor1
%   ----------------------Output------
-----------------------------------
%   H is homography Matrix
%   ----------------------Others
%   Yining Shen
%   Imperial College London
%   Electrical&Electron2c Eng2neering
%   1.April.2017
X1 = coor_out(1, :);
Y1 = coor_out(2,:);
X2 = coor_in(1,:);
Y2 = coor_in(2,:);
n = size(coor_in, 2);
A1 = [-[X2; Y2; ones(1,n)]; zeros(3,
n); X1.*X2; X1.*Y2; X1];
A2 = [zeros(3, n); -[X2; Y2;
ones(1,n)]; Y1.*X2; Y1.*Y2; Y1];
A=[];
for i=1:n
A=[A A1(:,i) A2(:,i)];
end
% Decomposition
[~, S, V] = svd(A');
% choose the smallest in U and take
the corresponding one in V.
H = (reshape(V(:,9), 3, 3)).';
end
```

Matching

```
function
[score,index]=fNearestMatch(color_hist
,color_hist2)
%color_hist & color_hist should be
N1*M and N2*M M=dimension N1,N1=
number or centers
%   Yining Shen
%   Imperial College London
%   Electrical&Electron2c Eng2neering
%   1.April.2017
mult = -2 * color_hist*color_hist2.' ;
% descriptor numbers x numbins
Ttrain_square = sum(color_hist.*
color_hist,2);          % numbins x 1
Ttest_square =
sum(color_hist2.*color_hist2,2); %
descriptor numbers x 1
a2plusb2 = bsxfun(@plus,
Ttrain_square,Ttest_square.');
%descriptor numbers x numbins
a2_2ab_b2 = bsxfun(@plus, a2plusb2,
mult );        % descriptor numbers x
numbins
[score, index] = min(a2_2ab_b2, [],
2);   % every raw stores all distance
of a descriptors to all centres
end
```

Find inliers using RANSAC

```
function [coord1,coord2] =
fFindInliers(pts1,pts2)
% [H corrPtIdx] =
findHomography(pts1,pts2)
%   Find the homography between two
planes using a set of corresponding
%   points. PTS1 =
[x1,x2,...;y1,y2,...]. RANSAC method
is used.
%   corrPtIdx is the indices of
inliers.
coef.minPtNum = 4;
coef.iterNum = 200;
coef.thDist = 10;
coef.thInlrRatio = .1;
[~,Idx] =
ransac1(pts1,pts2,coef,@fHomographyMat
rix,@calcDist);
coord1=pts1(:,Idx);
coord2=pts2(:,Idx);
end

function d = calcDist(H,pts1,pts2)
%   Project PTS1 to PTS3 using H, then
calcultate the distances between
%   PTS2 and PTS3
n = size(pts1,2);
pts3 = H*[pts1;ones(1,n)];
pts3 =
pts3(1:2,:)./repmat(pts3(3,:),2,1);
d = sum((pts2-pts3).^2,1);
end
```

Projection form image A to B using Homography

```
function y = fHomoTransform(H,image)
%   ----------------Input-----------
-------------------------------------
%   H - Homography matrix or
Fundamental Matrix
%   coor - Coordinates in Original
image
%   ----------------Output----------
-------------------------------------
%   A - After transform
%   ---------------Others-----------
%   Yining Shen
%   Imperial College London
%   Electrical&Electron2c Eng2neering
%   1.April.2017
coor=zeros(2,size(image,1)*size(image,
2));
count=1;
trans=zeros(size(image,1),size(image,2
));
for i=1:size(image,1)
    for j=1:size(image,2)
        temp=(H*[j;i;1]);
coor(:,count)=round([temp(1);temp(2)].
/temp(3));
       if
(coor(2,count)<=size(image,1) &&
coor(1,count)<=size(image,2) &&
coor(2,count)>0 && coor(1,count)>0)
%          if coor(1,count)<=0
%              coor(1,count)=1;end
%          if coor(2,count)<=0
%              coor(2,count)=1;end

trans(coor(2,count),coor(1,count))=ima
ge(i,j);
       end
        count=count+1;
    end
end
y=uint8(trans);
% end
% figure
% imshow(y)
end
```

Plot epiples and pipolar lines

```
function
fEpipoles(F1,image1,coord_1,coord_2)
% F1 --- fundamental matrix
% image1 --- original image
% coord_1/2 --- corresponding
coordinates pairs.
% Yining Shen
% Zhejiang University
% 03/04/2017
u=F1*[coord_1;ones(1,size(coord_1,2))]
;
kb=[-u(1,:)./u(2,:);-u(3,:)./u(2,:)];
figure
imshow(image1)
hold on
t=1:size(image1,2);
y=kb(1,:)'*t+kb(2,:)'*ones(1,size(imag
e1,2));
plot(t,y);
hold on
scatter(coord_2(1,:),coord_2(2,:))
end
```

Main scripts example of Q1 and Q2.1), including automatic correspondence or manually chosen points, estimating H, HA, projection, and Fundamental Matrix and epipolear lines etc.

```
%   ----------------------
%   Yining Shen
%   Imperial College London
%   Electrical&Electronic Engineering
%   5.April.2017
clear
clc
close all
initialize
%%%load image and zero padding
image1=rgb2gray(tsukuba_scene1);
% image1=boat3;
image1=imresize(image1,0.5);
image1=padarray(image1,[20 20]);
imshow(image1)
image2=rgb2gray(tsukuba_scene2);
% image2=boat3;
image2=imresize(image2,0.5);
image2=padarray(image2,[20 20]);
figure
imshow(image2)
HA_large=[];
% for iter=1:20
%%%find coner point
y1=fHarrisDetector(image1,20000000);
coordinate1=y1;
color_hist1=fGenerateDescriptor(y1,ima
ge1);
% coord_1 = ginput();


y2=fHarrisDetector(image2,20000000);
coordinate_temp=y2;
color_hist2=fGenerateDescriptor(y2,ima
ge2);
% coord_2 = ginput();

%% find nearest between image A and B
tic
[score,index]=fNearestMatch(color_hist
1,color_hist2);
toc

% Homography Matrix using RANSAC
y2_corr=y2(:,index);
[y1_corrPtIdx, y2_corrPtIdx] =
fFindInliers (y1,y2_corr);
H=fHomographyMatrix(y1_corrPtIdx,y2_co
rrPtIdx);
% original image
figure
imshow(image1)
hold on
scatter(y1(1,:),y1(2,:));
% projected image
tsugray1_hat =
fHomoTransform(H,image1);
figure
imshow(tsugray1_hat)
% estimate HA
leng=length(y1_corrPtIdx);
HA=zeros(1,leng-3);
for len=4:ceil(leng*0.8)
    len
    HA_large=[];

for iter=1:10
%%
num=randperm(leng);

    iter

H=fHomographyMatrix(y1_corrPtIdx(:,num
(1:len)),y2_corrPtIdx(:,num(1:len)));

y2_hat=H*[y1_corrPtIdx(:,num(len+1:end
));ones(1,leng-len)];
y2_hat=y2_hat./[y2_hat(3,:);y2_hat(3,:
);y2_hat(3,:)];
error=(round(y2_hat(1:2,:))-
round(y2_corrPtIdx(:,num(len+1:end))))
;
dist=sqrt(error(1,:).^2+error(2,:).^2)
;
if mean(dist)>0.1
HA_large=[HA_large mean(dist)];
end
end
HA(len-3)=mean(HA_large);
end

%% calculate FD mannually
F1=fFundamentalMatrix(coord_1,coord_2)
;
%% calculate FD automatically
y1=fHarrisDetector((image1),50000000);
color_hist1=fGenerateDescriptor(y1,ima
ge1);

y2=fHarrisDetector((image2),50000000);
```

```
color_hist2=fGenerateDescriptor(y2,ima
ge2);
[~,index]=fNearestMatch(color_hist1,co
lor_hist2);

y2_corr=y2(:,index);
y1_corr=y1;
% discard outliers using inliers
obtained before
F=fFundamentalMatrix(y1_corrPtIdx,y2_c
orrPtIdx);
fEpipoles(F3,image2,y1_corrPtIdx(:,1:1
0:end),y2_corrPtIdx(:,1:10:end));
fEpipoles(F,image2,coord_1,coord_2);
```

Calculating disparity map and depth map.

```
%   ----------------------
%   Yining Shen
%   Imperial College London
%   Electrical&Electron2c Eng2neering
%   5.April.2017
clc
clear
close all
% disparity map  %% small image 17,16/
16,15 large image 20,21
initialize
I1 = imresize(img20,0.25);
I2 = imresize(img21,0.25);
figure
A = stereoAnaglyph(I1, I2);
imshow(A);

range=[0 64];
d =
disparity(rgb2gray(I1),rgb2gray(I2),'B
lockSize',13,'DisparityRange',range);

figure
imshow(uint8(d),[]);% title('original
disparity');
axis image;
colorbar

%% depth map
pixelsize=0.014;
baseline=12.5*pixelsize;
focal_distance=22;
index=find(d>0);
d(index)=d(index)+0.1*randn(size(d(ind
ex),1),size(d(index),2)); % add noise

disparityMap=d;
z=focal_distance*baseline./(disparityM
ap*pixelsize);              % depth
formula
xl=ones(size(rgb2gray(I1),1),1)*[1:siz
e(rgb2gray(I1),2)];
yl=[1:size(rgb2gray(I1),1)]'*ones(1,si
ze(rgb2gray(I1),2));
x_depth=xl.*z/focal_distance;
y_depth=yl.*z/focal_distance;
figure
mesh(z);
```

Main scripts example of Q2.2), including estimating F, plotting epipolar lines and epipoles.

```
%   ----------------------
%   Yining Shen
%   Imperial College London
%   Electrical&Electron2c Eng2neering
%   5.April.2017
clear
clc
close all

initialize
%%%load image and zero padding
image1=mountain_1;
image1=imresize(image1,0.5);
image1=padarray(image1,[20 20]);
image1=rgb2gray(image1);
imshow(image1)
% coord_1=ginput();
image2=mountain_2;
image2=imresize(image2,0.5);
image2=padarray(image2,[20 20]);
image2=rgb2gray(image2);
% figure
% imshow(image2)
% coord_2=ginput();
%% calculate FD mannually
F1=fFundamentalMatrix(coord_1,coord_2)
;
%% calculate FD automatically
y1=fHarrisDetector((image1),50000000);
color_hist1=fGenerateDescriptor(y1,ima
ge1);
y2=fHarrisDetector((image2),50000000);
color_hist2=fGenerateDescriptor(y2,ima
ge2);
[~,index]=fNearestMatch(color_hist1,co
lor_hist2);
y2_corr=y2(:,index);
y1_corr=y1;
% using all corr points
F2=fFundamentalMatrix(y1_corr,y2_corr)
;
% discard outliers
[F3,indx]=estimateFundamentalMatrix(y1
_corr',y2_corr','method','RANSAC',...
'NumTrials',10000,'DistanceThreshold',
0.1); %build in function for
comparison
y1_corrPtIdx=y1_corr(:,indx);
y2_corrPtIdx=y2_corr(:,indx);
F=fFundamentalMatrix(y1_corrPtIdx,y2_c
orrPtIdx);
fEpipoles(F3,image2,y1_corrPtIdx(:,1:1
0:end),y2_corrPtIdx(:,1:10:end));
fEpipoles(F,image2,coord_1,coord_2);
fEpipoles(F3,image1,y2_corrPtIdx(:,1:1
0:end),y1_corrPtIdx(:,1:10:end));
fEpipoles(F,image1,coord_2,coord_1);
% fEpipoles(F2,image2,coord_1);
[U,S,V]=svd(F);
e(:,1)=V(:,3)/V(3,3);
e(:,2)=U(:,3)/U(3,3);
```

Appendix III

Some experiment results

Q1.1), Q1.2) a) Interesting points detection



(a)Gound truth interesting points



(b)scene1 interesting points



(a)threshold=3e7



(b) threshold=3e8



(a)NotreDamede_Paris1 interesting points



(b)boat interesting points

Fig.1 Interesting points detection

Q1.2) b) c) Descriptors and Matching

Using colour histogram descriptors



Fig.2 Tsukuba matching using colour histogram descriptors

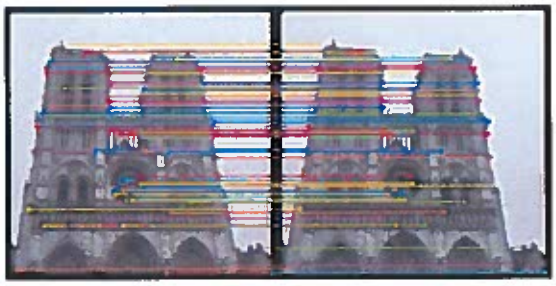


Fig.3 NotreDamede_Paris1 matching using colour histogram descriptors

Using SIFT descriptors





Fig.4 NotreDamede_Paris1 matching using SIFT descriptors

Both two methods shows similar results in matching which means colour histogram and SIFT are both suitable for this basic experiment.

Q1.3) a) b) c) & Q2.1) a) b) c) Some projection results



(a) (b) (c)

Fig.5 Projected Image using (a)automatic correspondences(b)carefully chosen correspondeces (c)less accuracy manually chosen correspondeces



(a) (b) (c)

Fig.6 Zoomed out Image Projection (a)Original Image (b)Zoomed out Image (c)Projection Image



(a) (b) (c)

Fig.7 Rotated Image Projection (a)Original Image (b) Rotated in Image (c)Projection Image



Fig.8 Rotated Image Projection



Fig.9 Slightly Affine



Fig.10 parallel translation

Q1.3) d) & Q2.2) a) b) Fundamental Matrix related



Fig.11 Epipoles if possible and epipolar lines for NotreDamede_Paris1



Fig.12 Epipoles if possible and epipolar lines for desk1



Fig.13 Epipoles if possible and epipolar lines for desk2

Q2.2) c) d) e) Disparity Map & Depth Map



(a)left+right image (b)disparity map (c) depth map

Fig.14 Laboratory scene



(a)left+right image (b)disparity map (c) depth map

Fig.15 Laboratory scene

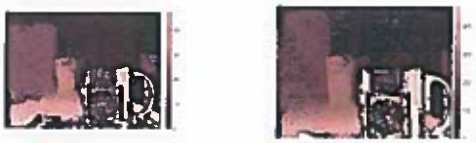


Fig.16 Disparity map of (a) original image and (b) zoomed in image



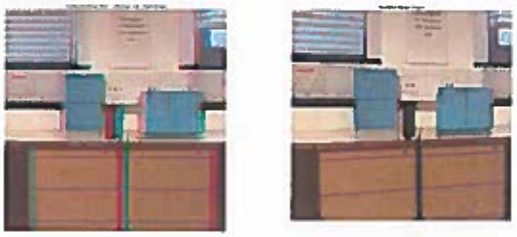
Fig.17 Depth map of (a) original image and (b) zoomed in image



Noise power=0 Noise power=0.01 Noise power=0.02

Fig.18 Noisy depth map

Q2.2) f) Rectification



(a)left+right image (b)rectification

Fig.19 laboratory



(a)left+right image (b)rectification

Fig.20 tsukuba